

Research

Assessing above ground biomass of Wunbaik Mangrove Forest in Myanmar using machine learning and remote sensing data

Win Sithu Maung¹ · Satoshi Tsuyuki² · Takuya Hiroshima² · San San Htay³

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Abstract

Estimating Above Ground Biomass (AGB) of mangroves provides crucial information for regional and global blue carbon strategies. Despite growing interest in mangrove AGB estimation using a remote sensing approach, the effectiveness of this advanced method has not been evaluated for mangrove forests in Myanmar. This study estimated the AGB of Wunbaik Mangrove Forest (WMF) in Myanmar using machine learning models with data from Sentinel-2, Sentinel-1, and Advanced Land Observing Satellite Phased Arrayed L-band Synthetic Aperture Radar (ALOS-PALSAR) HH and HV polarizations and Canopy Height Model. Reference AGB information was derived from field inventory plots via an allometric equation. Initially, we tested machine learning models such as Random Forest (RF), Support Vector Machine (SVM), and Extreme Gradient Boost (XGBoost). The results revealed that the RF model outperformed the other models, yielding a higher coefficient of determination (R^2) of 0.33 and a Root Mean Square Error (RMSE) of 32.05 Mg ha⁻¹. We then set up different scenarios to improve the performance of the RF model for AGB estimation. Through different feature selection approaches, we identified features highly correlated with the field AGB, enhancing the RF model's performance and resulting in the highest improvement in R^2 : 0.48 and RMSE: 28.12 Mg ha⁻¹. In this study, an AGB map of the WMF was generated using the best RF model. The predicted AGB distribution ranged from 22.266 Mg ha⁻¹ to 181.948 Mg ha⁻¹ in 2019. Compared with global datasets such as GEDI L4B and ESA CCI predictions, the proposed method provides a more accurate prediction of AGB for Wunbaik Mangrove, enhancing blue carbon information in Myanmar.

Keywords Mangrove forest · Above ground biomass · Above ground carbon · Remote sensing approach · Machine learning models · Feature selection

1 Introduction

To combat global climate change by reducing greenhouse gas (GHG) emissions into the atmosphere, the blue carbon strategy has become highly popular because it highlights the role of coastal ecosystems [1]. Mangrove forests function as one of the main blue carbon components, as do seagrass, salt marsh, and coral reefs [2]. Mangroves are highly efficient at capturing atmospheric carbon dioxide (CO₂) through photosynthesis and storing it in their biomass and sediments. Their dense

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✉ Satoshi Tsuyuki, tsuyuki@fr.a.u-tokyo.ac.jp; Win Sithu Maung, wins0323@jircas.go.jp; sithumg1989@gmail.com; Takuya Hiroshima, hiroshim@g.ecc.u-tokyo.ac.jp; San San Htay, sansanhtay43@gmail.com | ¹Social Sciences Division, Japan International Research Center for Agricultural Sciences (JIRCAS), Tsukuba Ibaraki 305-8686, Japan. ²Department of Global Agricultural Sciences, Graduate School of Agriculture and Life Sciences, The University of Tokyo, Tokyo 113-8654, Japan. ³Freelance Researcher, Yangon, Myanmar.



vegetation and extensive root systems trap large amounts of carbon, making them among the most effective terrestrial carbon sinks [3]. Mangroves can sequester carbon at rates up to four times higher than those of tropical rainforests, making them invaluable assets in global carbon cycling [3, 4]. Despite the substantial contribution of mangroves to carbon storage, mangrove forests worldwide are being degraded due to shrimp culture, paddy fields, urban development, and over-harvesting of fuelwood [5]. In Southeast Asia, mangrove deforestation between 2000 and 2012 was an average rate of 0.18% per year, and aquaculture and agricultural expansion were described as the major pressures on mangrove ecosystems [6, 7]. However, anthropogenic pressures vary depending on regional livelihood cultures, so regional studies are needed to customize local issues and seek solutions for protecting remaining mangroves and formulating effective restoration programs.

To mitigate GHG emissions to combat global climate change, Myanmar has joined the United Nations Framework Convention on Climate Change (UNFCCC), the Kyoto Protocol and REDD+ (Reducing Emission from Deforestation and forest degradation in Developing countries) programs. Despite such appreciable activities in the forestry sector, very few studies have explored mangrove carbon information in Myanmar. Mangrove AGB estimation contributes valuable information to blue carbon trends regionally and globally [8]. With the support of the REDD+ program, mangrove restoration programs can be enhanced by highlighting the carbon richness of mangrove forests [9]. Previous AGB studies have explored AGB analyses for mangroves in the Ayeyarwaddy and the Tanintharyi regions in Myanmar [10–13]. However, these studies are field inventory-based approaches that cover only sample plot areas. To achieve effective mangrove restoration programs, landscape-scale AGB information is necessary to highlight the carbon richness of a whole study area. For this purpose, relying on field surveys alone is impossible but can be accomplished through a remote sensing approach. Therefore, this study will also accommodate spatially estimated mangrove carbon information using advanced remote sensing and machine learning approaches.

With the development of technology, remote sensing is undoubtedly an effective approach for estimating mangrove forest carbon. Field observations for estimating the AGB of a whole mangrove area are costly, time-consuming, and laborious. Many studies have explored and developed several remote sensing approaches for monitoring mangrove AGB at a regional scale [14–16]. Some studies have also successfully applied LiDAR and UAV data for mangrove AGB estimation [17–19]. When selecting an effective method for mangrove AGB estimation, many studies have shown the superiority of machine learning approaches over traditional methods [20]. Machine learning models can easily integrate various data types, such as optical and SAR satellite imagery, UAV, LiDAR, and ancillary data (e.g., topography, canopy height, and climate), producing more accurate and precise AGB estimations. They can analyze and model nonlinear relationships between variables, which traditional methods often struggle to capture. In addition, the application of multispectral optical and Synthetic Aperture Radar (SAR) satellite imagery has contributed impressive accuracy in large-scale AGB estimation [20–22]. Nevertheless, the accuracy of AGB estimation varies depending on the effectiveness of input features and methods used [23]. Accordingly, integrating optical and SAR data with texture and canopy height information can improve the accuracy of mangrove AGB estimation.

The retrieval of accurately estimated AGB information also needs to be localized depending on the nature of the mangroves and their related environmental and anthropogenic factors. A remote sensing and machine learning approach was explored for estimating AGB of evergreen and deciduous forest types in Myanmar [24]. However, there is no information on AGB studies using remote sensing for mangrove areas in Myanmar, especially the Wundwin mangrove area. Considering these challenges and limitations, this study aims to contribute improved AGB information for the Wundwin mangrove area by combining advanced remote sensing and machine learning approaches with field-referenced AGB data, highlighting its carbon richness. We also delineate the integration of Earth Observation data using feature selection scenarios to improve the performance of the machine learning model.

2 Materials and methods

2.1 Study area

The study area is one of the largest mangrove ecosystems in Myanmar, with a land area of approximately 55,000 ha. It is located between 19°05' 00" and 19°25' 00" N, and 93° 50' 00" and 94° 05' 00" E in Rakhine coastal region (Fig. 1). The Forest Department of Myanmar managed mangrove forests as the Wundwin Reserved Mangrove Forest (WRMF) in 1931, the Mingyaung Public Mangrove Forest in 2007, and the Extended Mingyaung Public Mangrove Forest in 2018. The altitude in the study area ranges from 0 to 95 m above sea level (m a.s.l.). According to the Myanmar Information Management Unit (MIMU) (https://themimu.info/states_regions/rakhine), temperatures between 27 °C and 33 °C and 4860 mm of annual rainfall were recorded in the study area in 2019. Local populations heavily depend on mangrove forests for their livelihoods because of rapid population growth, low income, and scarcity of electricity. Large areas of mangrove forests

are cleared for their main livelihoods, including shrimp ponds and paddy fields [25, 26]. To compensate for mangrove loss, the Forest Department has implemented mangrove restoration projects since 2006. However, there are limitations in evaluating the progress of the projects and acquiring enough information for the management of Wunbaik Mangrove Forest since the study area is inaccessible and remote.

2.2 Field inventory data collection

This study used field inventory data collected from September 15 to October 10, 2019. During the field observations, 50 sample plots were established with a 10 m × 10 m square design using a handheld GPS receiver (GARMIN etrex 10) (Garmin Ltd., United States) with 3 m horizontal accuracy. Tree species and Girth at Breast Height (GBH) were collected within 100 m² of the sample plots with the aid of field staff of the Forest Department and local labor (Fig. 2). GBH values were converted into Diameter Breast Height (DBH) values by multiplying by 0.347. Since these 50 plots are located at naturally recovering sites where mangroves regenerated after 5 years of abandonment from shrimp farming [27], the mangrove stands of this field survey are growing with a small DBH.

Additionally, we utilized National Forest Inventory (NFI) plots collected by the Forest Department of Myanmar and the Food and Agricultural Organization (FAO) in 2017 [28]. Owing to the systematic sampling design of the NFI dataset, some plots are located in non-mangrove areas, such as shrimp ponds, paddy fields, and barren lands. Based on the LULC map from the previous study, we selected inventory plots that only fell within the closed mangrove areas classified as dense mangrove forests with minimal anthropogenic disturbance and canopy gaps [29]. The 38 selected NFI plots were integrated with the former 50 inventory plots at recovering sites. In total, 88 field plots covering different statuses of naturally recovering and closed mangroves were selected for this AGB estimation.

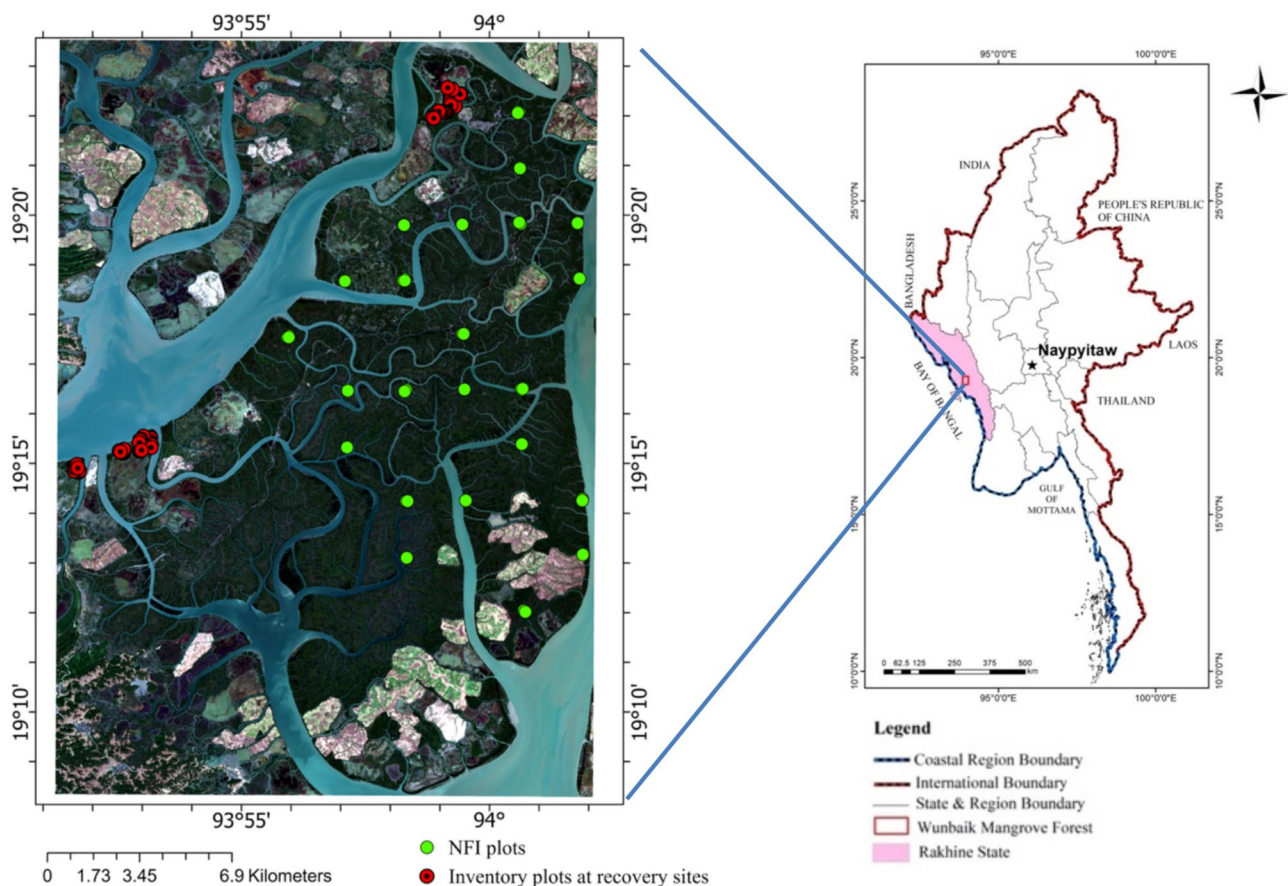


Fig. 1 Location of the study area, Wunbaik Mangrove Forest in Myanmar (Sentinel 2 true color image from 2019)

2.3 Field biomass calculation using allometric equations

To calculate mangrove AGB, we chose an allometric equation suitable for the study area since it was developed by Komiyama for Southeast Asian mangroves [30]. Wood density values were obtained from reliable resources (Global Wood Density Dataset and mangrove studies in Myanmar [10] and especially Southeast Asia). As per Eq. (1), DBH and wood density were used to calculate the plot-based AGB for the 88 sampling plots.

$$AGB = 0.251\rho(DBH)^{2.46} \quad (1)$$

where AGB is above ground biomass in $Mg\ ha^{-1}$, ρ represents the wood density value of each mangrove species, and DBH represents the diameter at the height of all the mangrove species in each sampling plot.

Among the total 88 field plots, 50 plots were located at the recovering sites, and 38 plots were selected from the NFI dataset. The AGB values of the 50 field inventory plots at the recovering sites are $100\ m^2$ and can be directly matched with the 10 m spatial resolution of satellite images. Since the plot size of the NFI plots is $0.08\ ha\ (800\ m^2)$, the AGB values of the NFI plots were divided by 8 to match the same scale as the $100\ m^2$ field inventory plots at the recovering sites and the 10 m resolution of the satellite imagery.

After AGB information at the landscape scale was predicted, the Above Ground Carbon (AGC) content in $Mg\ C\ ha^{-1}$ was calculated using the following empirical equation (Eq. 2) published in the Indonesian National Standard (SNI 7724:2011) [31–33].

$$AGC = AGB * 0.47 \quad (2)$$

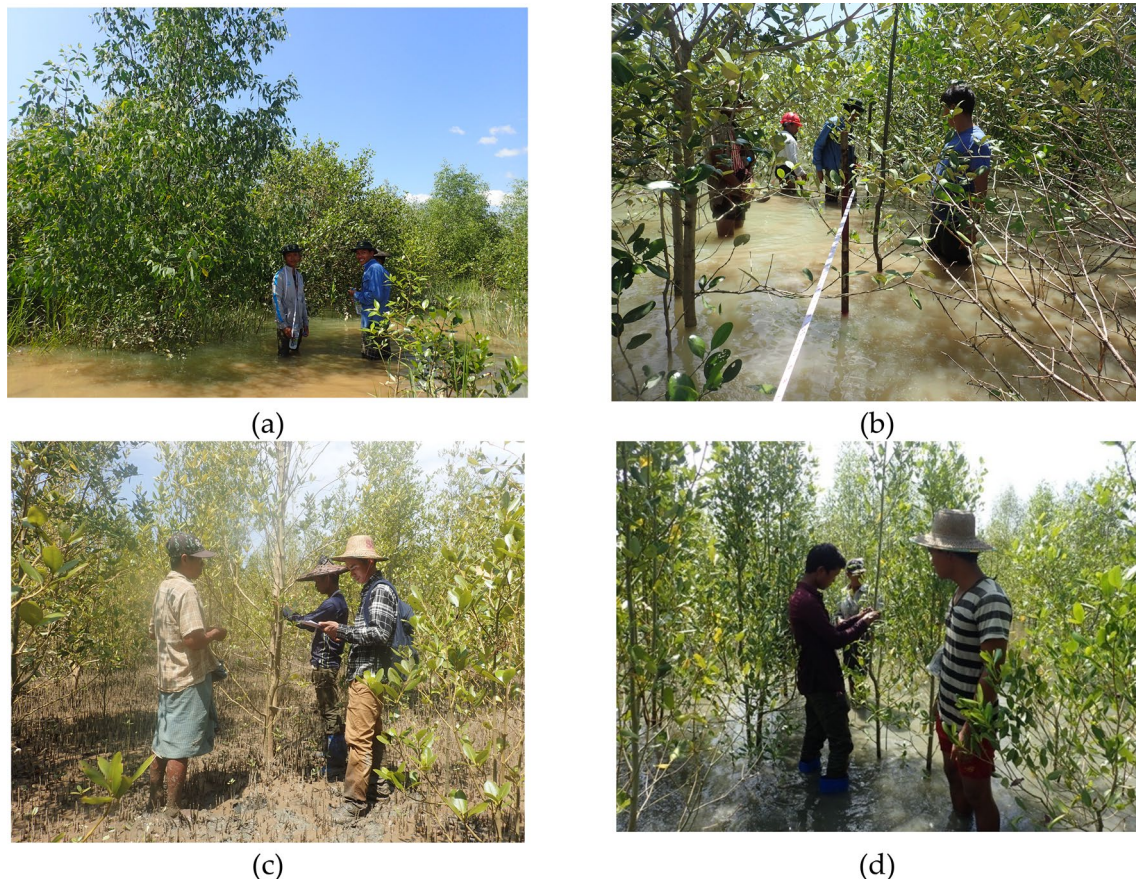


Fig. 2 Mangrove field survey in 2019 **a** mangrove forest in the sample plot area, **b** demarcating the sample plot, **c** and **d** recording mangrove specifications

2.4 Remote sensing data

2.4.1 Sentinel image

The study mainly used remote sensing information acquired from different sources of both optical and SAR sensors operated by the European Space Agency (ESA). Sentinel-2 optical sensor provides 13 multiple spectral bands with different versions of Level-1 C (orthorectified Top-of-Atmospheric reflectance) and Level-2 A (orthorectified Bottom-of-Atmospheric reflectance). The Sentinel-2 bands were downloaded from the Copernicus scientific data hub using the semiautomatic classification plugin (SCP) in QGIS software. We utilized remote sensing information acquired from different sources of both optical and SAR sensors. Among the 13 multispectral bands offered by Sentinel-2, 10 bands with 10–20 m resolution were utilized for AGB estimation. The selected bands with a spatial resolution of 20 m were resampled to 10 m using the Nearest Neighbor method to ensure consistency with other bands. Additionally, various indices of Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), Soil Adjusted Vegetation Index (SAVI), Ratio Vegetation Index (RVI), Green NDVI (GNDVI), Inverted Red-Edge Vegetation Index (IRECI), Modified Chlorophyll Absorption in Reflectance Index (MCARI), Difference Vegetation Index (DVI), The Second Enhanced Vegetation Index (EVI-2). The details of each index were mentioned in Table 1.

Sentinel-1 images are captured by the SAR satellite equipped with C-band radar, which uses two modes of VV (vertical transmitting-vertical receiving) and VH (vertical transmitting-horizontal receiving) polarization, which are capable of imaging in cloudy and rainy situations with 10 m spatial resolution. From the Sentinel-1 image, VV and VH polarizations were also acquired from the Copernicus scientific data hub by selecting the dates that matched the field AGB data collection period. Sentinel-1 data were then calibrated, terrain corrected, and resampled into a 10 × 10 m² pixel size in SNAP (Sentinel Application Platform) software. In addition to the VV and VH bands, 5 SAR band transformations were also applied in AGB estimation. To explore the contribution of Sentinel-1 SAR data, other than the VV and VH bands, five SAR feature transformations (mentioned in Table 1) were also applied in this analysis. Additionally, eight texture features from the VV and VH bands were added to the dataset.

2.4.2 ALOS PALSAR-2

The study also utilized the global 25 m PALSAR/PALSAR-2 mosaic, a comprehensive SAR image compilation created by seamlessly mosaicking SAR imagery strips from the PALSAR/PALSAR-2 satellites. The SAR imagery was provided by the Japan Aerospace Exploration Agency (JAXA) in dual-polarization mode (HH/HV polarizations). JAXA conducted orthorectification and slope correction of the SAR imagery using the 90 m SRTM Digital Elevation Model. The pre-processed SAR bands for the year 2019 were downloaded using Google Earth Engine (https://developers.google.com/earth-engine/datasets/catalog/JAXA_ALOS_PALSAR_YEARLY_SAR) and resampled to a 10 m resolution using the Nearest Neighbor method. The images were exported into Tiff format and visualized in ArcGIS Pro. After matching with other images, feature information was extracted for AGB analysis.

2.4.3 Canopy height information

Among various forest attributes, canopy height information is one of the key factors that is highly correlated with forest AGB. The integration of canopy height information has been explored to improve the accuracy of forest AGB estimation. The use of canopy height information in LULC classification could improve the accuracy of classifying intricate mangrove ecosystems [29]. To improve the model performance in AGB regression, this study utilized Global Canopy Height Model (GCHM) datasets (<https://langnico.github.io/globalcanopyheight/>, accessed on 17 July 2023). The different canopy height information of the mangrove forests in this study area can help highlight and correlate mangrove AGB information. The GCHM was developed by fusing the Global Ecosystem Dynamic Investigation (GEDI) space-borne LiDAR mission and Sentinel-2 imagery [34, 35]. The GCHM dataset is available on the Google Earth Engine with a spatial resolution of 10 m.

Table 1 Description of features used in AGB analysis

Main Sensor/Source	Features	Central Wavelength/Formula	References
Sentinel-2	B2-Blue	490	[36]
	B3-Green	560	
	B4-Red	655	
	B5-Red Edge1	705	
	B6-Red Edge2	740	
	B7-Red Edge3	783	[37]
	B8-Near InfraRed (NIR)	842	
	B8A-Narrow NIR	865	
	B11-Short Wave Infra-Red) (SWIR-1)	1610	
	B12-Short Wave Infra-Red) (SWIR-2)	2190	
	Normalized Difference Vegetation Index (NDVI)	$\frac{\text{NIR}-\text{RED}}{\text{GREEN}+\text{NIR}}$	[38]
	Normalized Difference Water Index (NDWI)	$\frac{\text{GREEN}-\text{NIR}}{\text{GREEN}+\text{NIR}}$	
	Soil Adjusted Vegetation Index (SAVI)	$(1 + L) \left(\frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}+L} \right) L = 0.5$	[39]
	Ratio Vegetation Index (RVI)	$\frac{\text{NIR}}{\text{RED}}$	
	Green NDVI (GNDVI)	$\frac{\text{NIR}-\text{GREEN}}{\text{NIR}+\text{GREEN}}$	[40]
	Inverted Red-Edge Vegetation Index (IRECI)	$\frac{\text{Red Edge}(B7)-\text{Red}}{\text{Red Edge}(B5)}$	[41]
	Modified Chlorophyll Absorption in Reflectance Index (MCARI)	$\frac{\text{Red Edge}(B5)}{\text{Red Edge}(B6)}$	[42]
	Difference Vegetation Index (DVI)	$(\text{Red Edge}(B5) - \text{Red}) - 0.2 * (\text{Red Edge}(B5) - \text{Green}) * (\text{Red Edge}(B5) - \text{NIR})$	
	The Second Enhanced Vegetation Index (EVI-2)	$\text{NIR} - \text{RED}$	[43]
		$2.5 \left(\frac{\text{NIR}-\text{RED}}{\text{NIR}+2.4*\text{RED}+1} \right)$	[44]
Sentinel-1 SAR	VH	Horizontally polarized backscatter	[45]
	VV	Vertically polarized backscatter	
	VV-VH	SAR transformation	[46]
	VH-VV		
	VH/VV		
ALOS PALSAR-2	(VW + VH)/2	8 Texture Features of GLCM derived from VH	[47]
	Contrast/Dissimilarity/Homogeneity/ASM/Entropy/Mean/Variance/Correlation	8 Texture Features of GLCM derived from VV	
	HH	Horizontally polarized backscatter	[48]
Global Canopy Height Model	HV	Vertically polarized backscatter	
	CHM	Canopy height information	[34]

2.5 Machine learning models

2.5.1 Random forest

The Random Forest (RF) model is a machine learning algorithm and is now widely applied in both classification and regression because of its robustness to overfitting [49]. The RF model is an ensemble learning approach that integrates the predictions of multiple decision trees to improve the accuracy and robustness of predictions. Each decision tree in the ensemble is constructed using a subset of the training data and features, and the final prediction is obtained by averaging the predictions of all trees in the forest. In the RF algorithm, bootstrapped sampling is applied to extract a set of sample data from the training dataset. Using the bootstrapped samples and randomly selected features, each decision tree in the RF ensemble is constructed recursively. At each node of the tree, the algorithm selects the best split based on a criterion such as the mean squared error (for regression tasks) or Gini impurity (for classification tasks).

The performance of the RF model for different types of forest classification has been shown using Sentinel-2 imagery in Mediterranean environments [50]. The RF model has been utilized for LULC analysis using the Landsat multitemporal images from the Google Earth Engine (GEE) from 2001 to 2018 in North Korea [51]. Additionally, in forest AGB estimation studies, the RF algorithm has outperformed other methods with different remote sensing data [23, 52, 53].

2.5.2 Support vector machine

Support Vector Machine (SVM) is a supervised machine learning algorithm commonly used for classification and regression tasks. The objective of SVM is to find the optimal hyperplane that best separates data points belonging to different classes or predicts a continuous target variable. This hyperplane is determined by maximizing the margin, which is the distance between the hyperplane and the closest data points from each class, also known as support vectors.

SVM works by transforming the input data into a higher-dimensional feature space where the data points become linearly separable. In this transformed feature space, SVM seeks to find the hyperplane that separates the data points into different classes while maximizing the margin. The optimization process involves solving a convex quadratic programming problem to find the coefficients of the hyperplane that minimize a cost function, which penalizes misclassifications and encourages maximizing the margin. SVM is widely used in various applications, including image classification [54], tree species classification [55], and forest AGB estimation [56], due to its robustness and flexibility. The superior performance of SVM over other machine learning models in land cover classification was shown using Sentinel-2 imagery [57]. The SVM also contributed to improving the accuracy of Calabrian pine AGB estimation [58].

2.5.3 Extreme gradient boosting

Extreme Gradient Boosting (XGBoost) is an advanced machine learning algorithm [59] that works by sequentially building an ensemble of decision trees, where each subsequent tree corrects the errors made by the previous trees. XGBoost is designed to optimize both computational speed and model performance. It achieves this by employing a technique called gradient boosting, where each tree is trained on the residuals (the differences between the actual and predicted values) of the previous trees, gradually reducing the error. Additionally, XGBoost incorporates regularization techniques to prevent overfitting and improve the generalizability to unseen data.

XGBoost has gained popularity in various fields, including image classification and regression, and often outperforms other machine learning algorithms. XGBoost was found to be better than ANN for PolSAR image classification [60]. XGBoost was also utilized for mangrove classification using UAV hyperspectral and LiDAR data [61]. In the case of mangrove AGB estimation, XGBoost with a genetic algorithm feature selection approach outperformed other machine learning models [62].

2.5.4 Model evaluation

This study separated the dataset into 80% for training and 20% for testing. Machine learning models were trained for AGB estimation using the training portion. The testing dataset was used to evaluate the models' performance. For model evaluation, two well-known matrices, the coefficient of determination (R^2) and the root mean square error (RMSE), were applied in this study.

2.5.5 Workflow of AGB estimation

The workflow of AGB estimation is described in Fig. 3. The details are explained step by step below.

1. Image acquisition and preprocessing: Satellite image tiles of Sentinel-1, Sentinel-2, ALOS PALSAR, and the CHM were obtained from the multiple sources mentioned in Sect. 2.4. The downloaded tiles are georeferenced and mosaiced to cover the study area via ArcGIS Pro software. The study area was then extracted from the single mosaiced tile of each satellite product.
2. Index calculation and image stacking: Using related bands of Sentinel-1 and 2 images, the indices and SAR transformation described in Table 1 were calculated. After all necessary bands and indices were processed, the images were stacked together to form a comprehensive dataset consisting of 45 features for each pixel, which were later used as input variables in the machine learning models.
3. Plot-based feature extraction and dataset preparation: To create a dataset with explanatory and target variables, all 45 features were extracted using the latitude and longitude of 88 field inventory plots with field AGB values. The dataset was then saved into a csv format for analysis via machine learning regression. This study used 80% of the training dataset (60 plots) and 20% of the testing dataset (28 plots) for the mangrove AGB analysis.

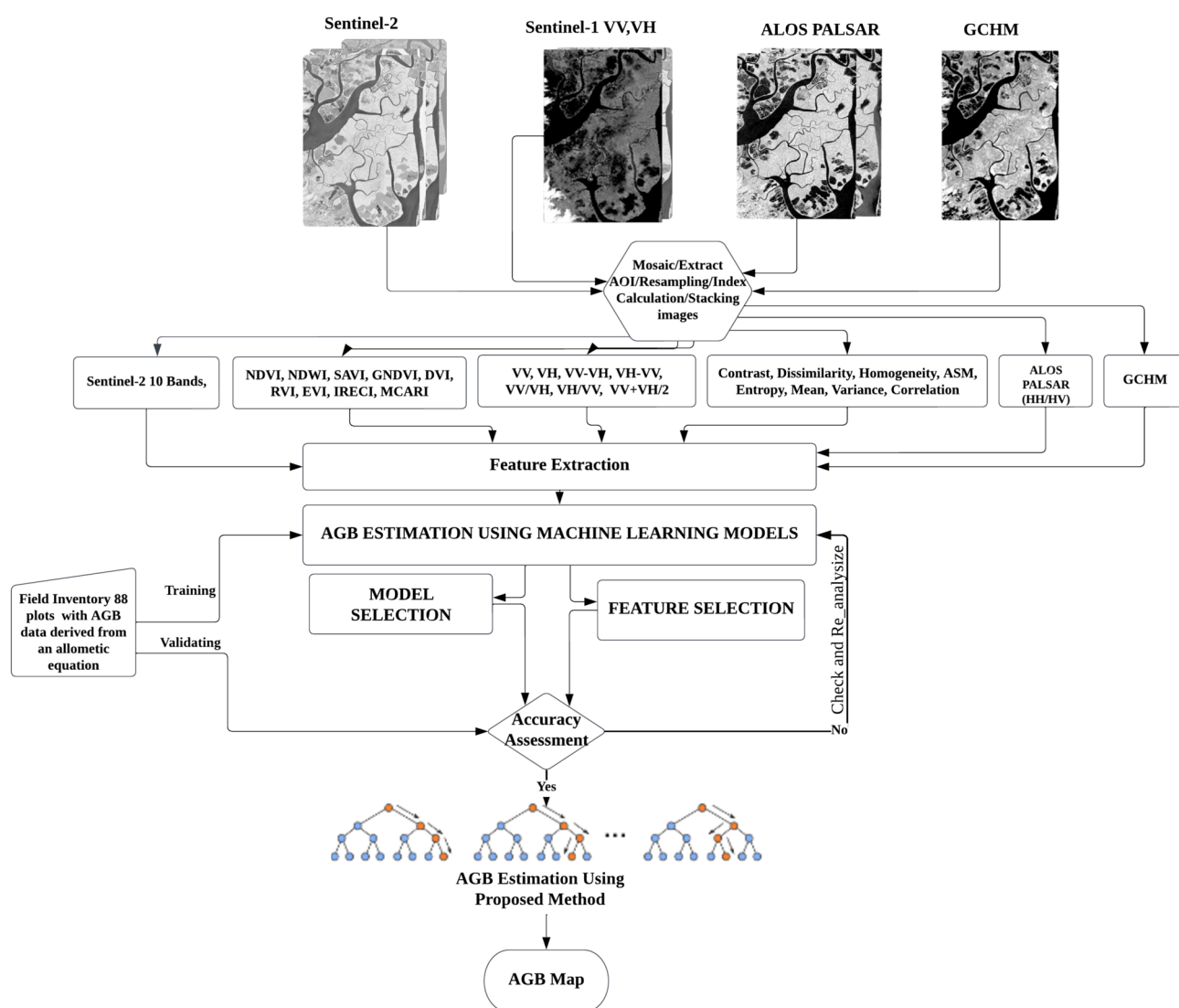


Fig. 3 Flowchart of mangrove AGB estimation in this study

4. **Model selection:** In this study, three different machine learning models, RF, XGBoost, and SVM, were initially tested to choose the most suitable model for AGB estimation in the Wunbaik mangrove area. This study used a grid search approach for model selection in which four parameters, namely, the number of trees (`n_estimators`), maximum depth of trees (`max_depth`), minimum samples leaf (`min_samples_leaf`), and maximum features (`max_features`), are tuned. The models were then trained using the training portion and evaluated via the testing data. After the model performance was evaluated via the unseen testing dataset, the model with the highest R^2 and lowest RMSE was selected for further analysis.
5. **Feature selection:** Different scenarios with various feature sets were explored to improve model performance. The aim was to identify the most relevant features for AGB estimation, and these features were selected based on the R^2 and RMSE values. The feature selection process helped narrow down the optimal set of features that provided the best performance in terms of accuracy and efficiency.
6. **AGB estimation at the landscape scale:** The best-performing machine learning model, with the selected feature set, was applied across the entire study area to estimate AGB at the landscape scale. The estimated AGB values were mapped across the study area and exported as a GeoTIFF file for comparison with existing global datasets.

2.5.6 Comparison with global AGB datasets

After exporting the AGB map using selected machine learning model and features, we compared our results with the global AGB datasets of the Global Ecosystem Dynamics Investigation (GEDI) L4B and the ESA Climate Change Initiative (CCI). The GEDI L4B product provides high-resolution estimates of above-ground biomass density (AGBD), which can be freely accessible on the NASA EARTHDATA website (https://daac.ornl.gov/GEDI/guides/GEDI_L4B_Gridded_Biomass.html). The GEDI L4B product is generated by integrating LiDAR waveform data collected from the GEDI instrument on the International Space Station with ancillary data. GEDI L4B data have a spatial resolution of 1 km and use waveform LiDAR to directly measure the vertical structure of vegetation, offering detailed biomass estimates through advanced algorithms that interpret the returned light signals from forest canopies.

The ESA CCI dataset, on the other hand, is derived from a combination of satellite observations, including radar and optical data, along with ground-based measurements to provide consistent and accurate AGB estimates across different ecosystems. This dataset has a spatial resolution of 100 m and integrates various sources of remote sensing data, such as Sentinel-1 SAR and optical data from Landsat. It estimates biomass using machine learning algorithms that correlate satellite signals with ground-truth biomass data. It can be downloaded from the European Space Agency website (<https://climate.esa.int/en/projects/biomass/>).

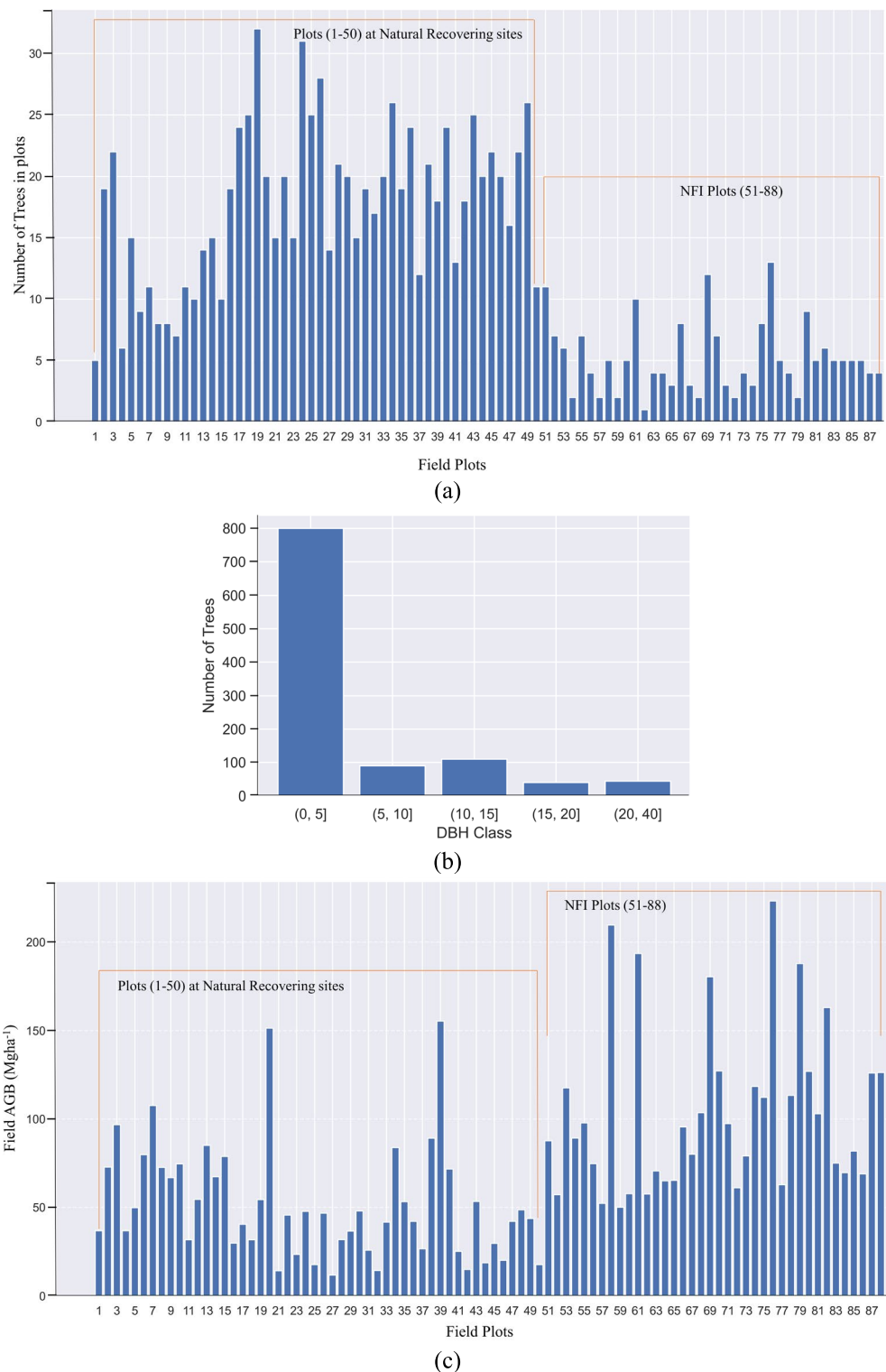
3 Results

3.1 Characteristics of the Wunbaik Mangrove Forest

As mentioned in Sect. 2.2, 50 plots were naturally recovering sites, and 38 plots were selected from NFI data. The 50 original plots from naturally recovering sites experienced recent disturbances, leading to ongoing changes in vegetation structure and composition. Owing to their rapid regrowth conditions, these recovering sites feature a greater number of young mangrove trees with small DBH values, resulting in lower AGB values, as presented in Fig. 4. In contrast, the 38 NFI plots from closed sites were characterized by stable, mature ecosystems. These sites have older mangrove trees with larger diameters, contributing to higher AGB values with a more uniform biomass distribution because of the stable and undisturbed nature of the ecosystem.

Among the total 88 inventory plots, 16 different mangrove species were found, and the total number of tree stands was 1084 trees, with a mean value of 12 trees per plot Fig. 4a. Among these, 13 species were found in the naturally recovering plots, whereas 14 species were identified in the NFI plots. Notably, *Avicennia marina* was the most dominant species in the recovering sites, with 517 trees, whereas the NFI plots featured a significant presence of *Xylocarpus granatum*, with 38 trees (S1). The observed DBH values ranged from 1 to 39.4 cm, and the details of the DBH class and number of tree stands are described in Fig. 4b. According to field-based AGB calculations, the field AGB values of the 88 plots ranged from 11.79 to 223.24 Mg ha⁻¹, with a mean AGB of 73.79 Mg ha⁻¹. The distribution of the field AGB is shown in Fig. 4c.

Fig. 4 **a** Number of trees observed in each plot, **b** DBH class belonging to a number of trees, and **c** field AGB results of inventory plots



3.2 Machine learning model selection

To select the most appropriate machine learning model for AGB analysis, three machine learning models of Random Forest, Support Vector Machine, and XGBoost, were preliminarily tested using all the features of the multisource remote sensing data. The results revealed that among the three different models, the RF model yielded higher R^2

values (0.33) and RMSE values (32.05 Mg ha^{-1}). SVM and XGBoost models produced R^2 values of 0.07 and -0.28 and RMSE values of 37.70 Mg ha^{-1} and 44.22 Mg ha^{-1} , respectively (Fig. 5). According to the grid search approach, the best hyperparameters for the RF model are 'max_depth': 5, 'max_features': 'log2', 'min_samples_leaf': 1, and 'n_estimators': 100. Therefore, the RF model was selected to estimate the AGB via a feature selection approach.

3.3 AGB estimation using feature selection

This study utilized a total of 45 features from multisource remote sensing Sentinel-2, Sentinel-1, ALOS-PALSAR, and CHM data. In the feature selection section, six different scenarios of feature combination were set up to improve RF estimation (Table 2). Scenarios 1–4 were created using features from Sentinel-1 and Sentinel-2, whereas scenario 5 included all the features from the multisensor data mentioned in Table 1. For scenario 6, features that were highly correlated with the field AGB values were chosen. We first performed a correlation analysis using all the variables and then selected the 14 features with correlation values greater than 0.3 and less than -0.4 for scenario 6 (Fig. 6). Among the different experiments of feature selection, a significant improvement was found in scenario 6 when the selected variables were highly correlated with the field AGB value. The RF model was leveraged using those 14 features with improved R^2 values of 0.48 and RMSE values of 28.22 Mg ha^{-1} on the test dataset, which are in bold in Table 2. For the whole dataset, the RF model produced an R^2 of 0.77 and an RMSE of 25.22 Mg ha^{-1} (Fig. 7).

Using the scikit-learn package in python, the features were also ranked to determine their importance in AGB estimation. Among the 14 selected features, NDWI and ALOS PALSAR were the most important features for AGB estimation, followed by GNDVI and RVI. The ranks of all 14 selected features with their Feature Important Values (FIV) is shown in Fig. 8.

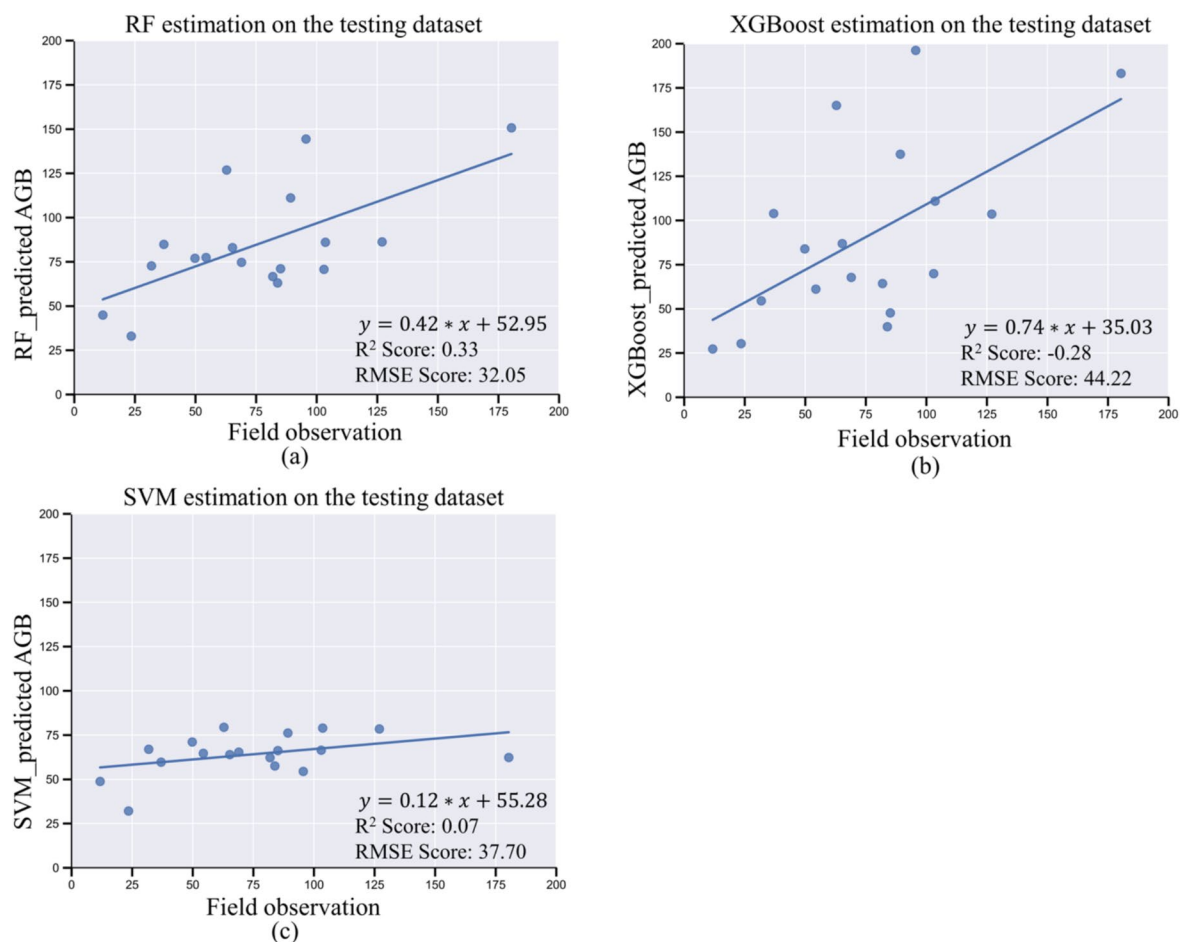
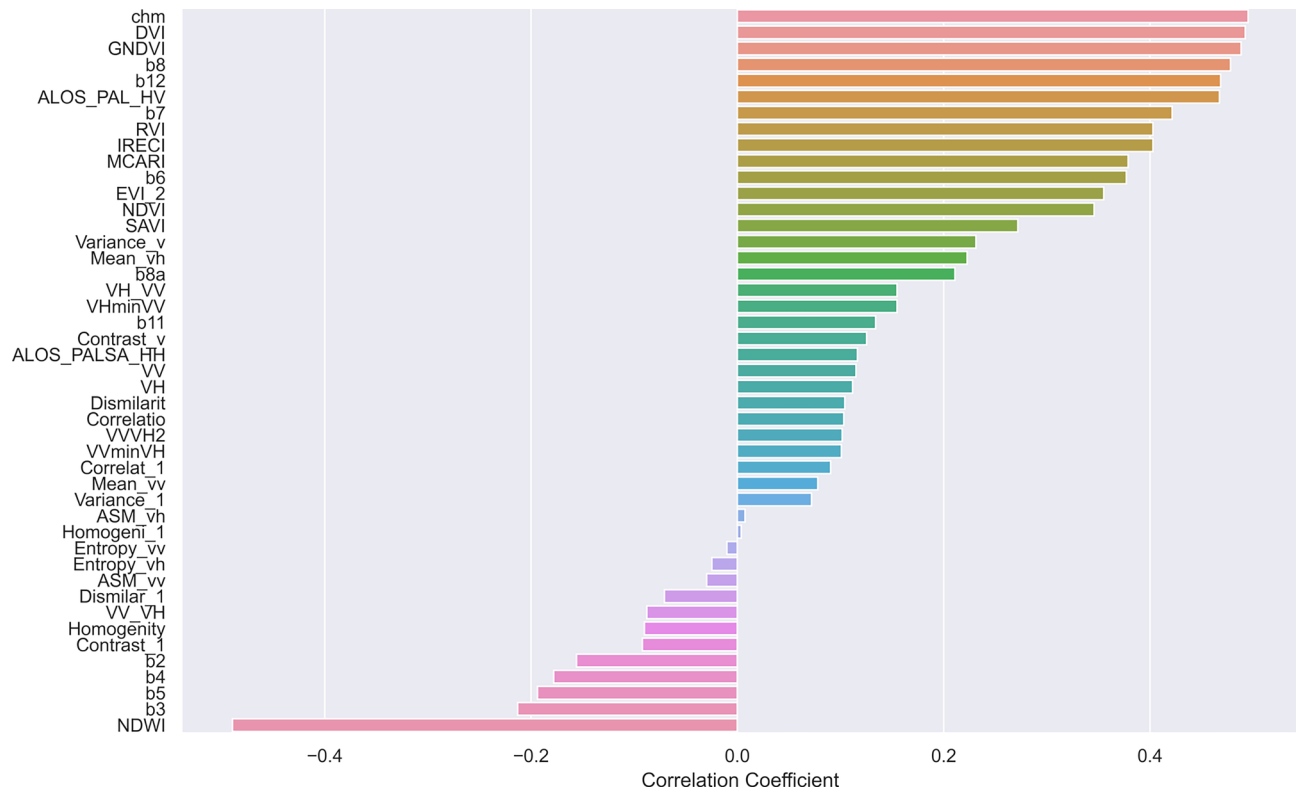


Fig. 5 Comparison of the predictions of **a** RF, **b** XGBoost, and **c** SVM for model selection

Table 2 Different feature selection scenarios used for AGB estimation to optimize RF model performance

Scenario	Features	Number of features	R ²	RMSE
1	Sentinel 2 Bands and VIs	19	0.34	31.79
2	Sentinel-1 VV/VH and transformation	7	−0.16	42.06
3	Sentinel 1/2	26	0.41	30.19
4	Sentinel 1/2 and texture features	42	0.37	31.08
5	All features	45	0.33	32.05
6	Selected features	14	0.48	28.22

**Fig. 6** Correlation values of all the remote sensing features with the field-observed AGB values

3.4 AGB and AGC maps of the Wunbaik Mangrove Forest

Since the RF model with 14 selected features produced the highest R² and RMSE values among the different scenarios, this study predicted the AGB information of the whole study area. The predicted AGB of the Wunbaik mangrove ranged from 22.27 to 181.95 Mg ha^{−1}, with a mean value of 88.79 Mg ha^{−1}. The AGB information was then converted into a carbon stock using the AGC conversion formula. The AGC of the Wunbaik mangrove ranged from 10.45 to 85.52 Mg C ha^{−1}, with a mean value of 41.73 Mg C ha^{−1}. The predicted results were exported into GeoTiff format and visualized in ArcGIS Pro. The distributions of the AGB and AGC information are also described in Fig. 9.

3.5 Comparison of Wunbaik mangrove AGB information with global datasets

This study compared mangrove AGB values with those of other global AGB datasets. In Fig. 10, the distribution of mangrove AGB values in this study was also presented by comparing them to the GEDI L4B and ESA_CCI datasets. The range of AGB values in this study (22.27–181.95 Mg ha^{−1}) more accurately reflects the observed variability within the Wunbaik

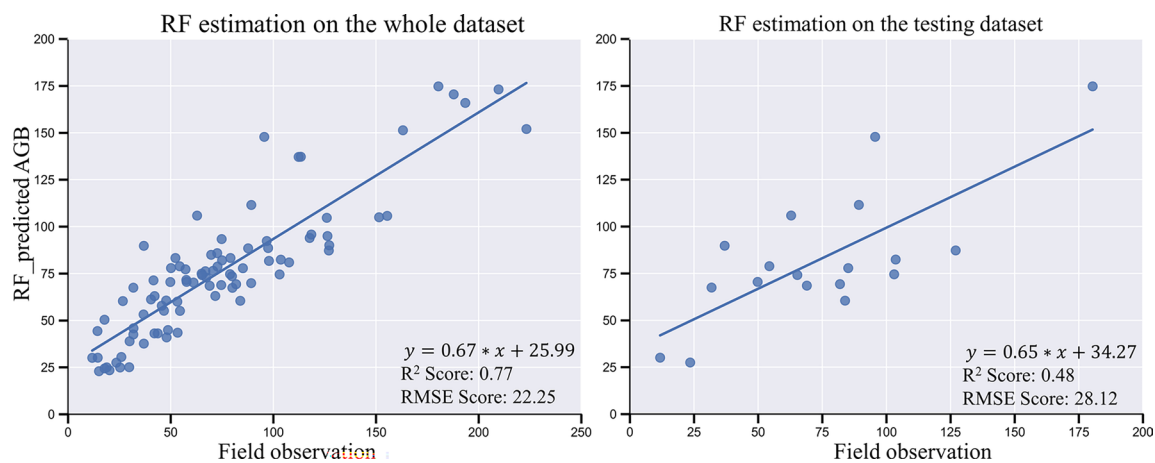
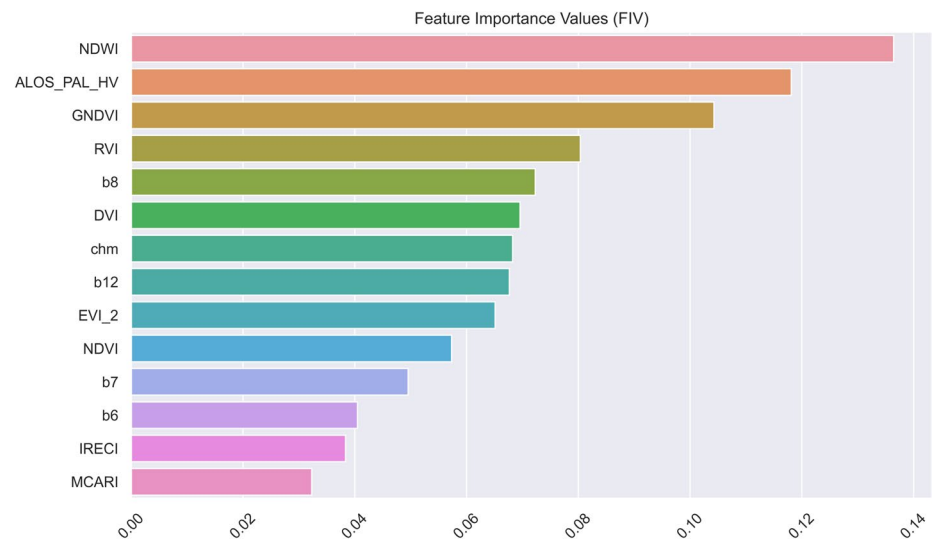


Fig. 7 AGB estimation using RF model with feature selection scenario 6 on the whole dataset and testing dataset

Fig. 8 Important feature ranking in this study



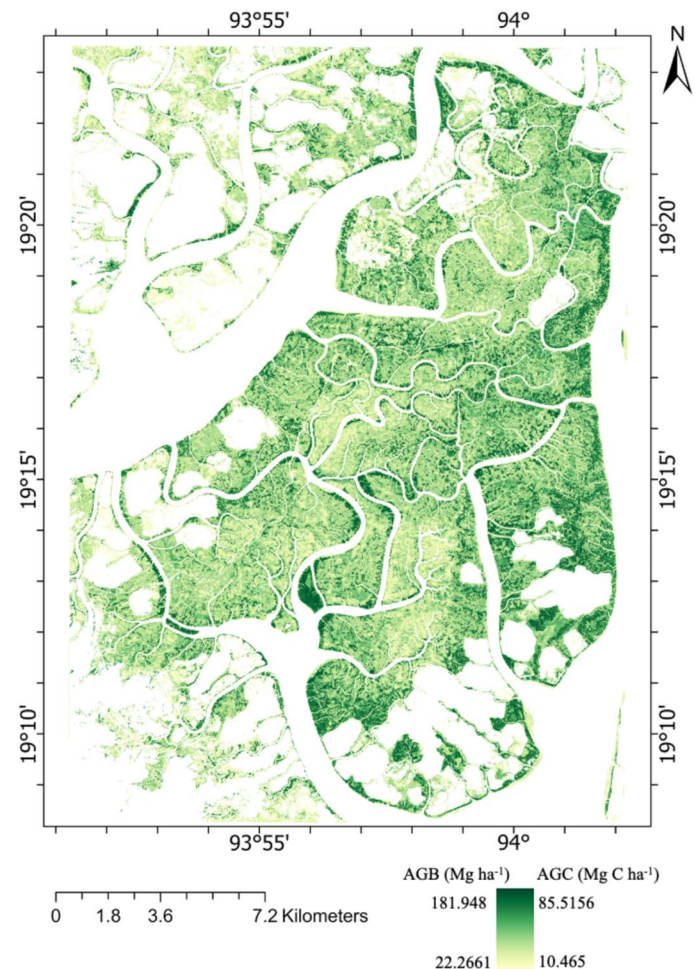
Mangrove Area. In contrast, the GEDI L4B and ESA CCI values are constrained to narrower values, with underestimated ranges of 0–60.17 Mg ha⁻¹ and 0–127.00 Mg ha⁻¹, respectively. The AGB values from this study closely aligned with the field-observed data, with a mean value of 88.79 Mg ha⁻¹, whereas the field mean was 73.79 Mg ha⁻¹. This contrasts sharply with the mean AGB values from GEDI L4B and ESA CCI, which are significantly lower at 11.60 Mg ha⁻¹ and 21.56 Mg ha⁻¹, respectively (Fig. 11).

4 Discussion

4.1 Effectiveness of the RS data for AGB estimation

This study attempted to estimate mangrove AGB using machine learning models with multisource remote sensing data for the Wunbaik mangrove area in Myanmar. Although many studies have explored mangrove AGB analysis using remote sensing, the contributions of RS data and models to AGB retrieval vary on the nature of mangroves, such as their growth, anthropogenic pressure, and hydrology. In this study, among several bands and indices derived from Sentinel-2 images, Bands 6, 7, 8 and 12, NDVI, NDWI, RVI, GNDVI, IRECI, MACRI, and EVI-2 were highly correlated with the mangrove AGB data of the field sample plots. Notably, NDWI had the greatest contribution to AGB prediction, with the highest FIV. It was assumed that 50 inventory plots were at recovering sites affected by human disturbances,

Fig. 9 Estimated AGB and AGC map using the RF model with selected features in Scenario 6 of this study



such as shrimp ponds and paddy fields. The tidal inundation at those plots was more frequent, so the water surface was more visible than that at the NFI plots.

Additionally, ALOS PALSAR HV also showed its importance in the AGB estimation in this study. Cross-polarized backscatters of HV polarization penetrate deeper into the vegetation canopy and interact with multiple scattering surfaces, such as branches, leaves, and trunks, which are critical for estimating biomass. This finding is consistent with other studies of mangrove AGB estimation. The Southeast Asian mangrove AGB dataset was developed using ALOS PALSAR [63]. Compared with HH bands, ALOS PALSAR HV had a greater FIV for estimating the AGB of mangrove forests in Vietnam [21]. In contrast, HH polarization primarily interacts with surface structures, making it less sensitive to the vertical complexity of mangrove forests. Moreover, when 7 features from the Sentinel-1 VV and VH bands and their transformations were used, they performed poorly, yielding the lowest R^2 (-0.16) and the highest RMSE (42.06 Mg ha^{-1}). This highlights the limited effectiveness of Sentinel-1 data alone for AGB estimation using RF model in this study, which is consistent with findings in Indonesia. [33]. We hypothesize that the longer-wavelength SAR data from ALOS-PALSAR are more effective for mangrove AGB estimation than the shorter-wavelength data from Sentinel-1. However, combining Sentinel-1 and Sentinel-2 features in Scenario 3 significantly improved the model, with an R^2 of 0.41 and an RMSE of 30.19 Mg ha^{-1} . Moreover, it is worth mentioning that the correlation between the CHM and field-observed AGB was the highest among the other variables. Therefore, the effectiveness of canopy height information is not negligible in predicting mangrove AGB in this study. In the case of Scenario 6, the use of 14 highly correlated features demonstrated the best performance among all the scenarios, with an R^2 of 0.48 and an RMSE of 28.22 Mg ha^{-1} . This approach specifically targeted variables that had the strongest correlation with the field AGB values. We suggest that careful feature selection is more effective than merely increasing the number of features. By eliminating less relevant or redundant data, the model was able to focus on the most informative variables, thus improving its predictive accuracy.

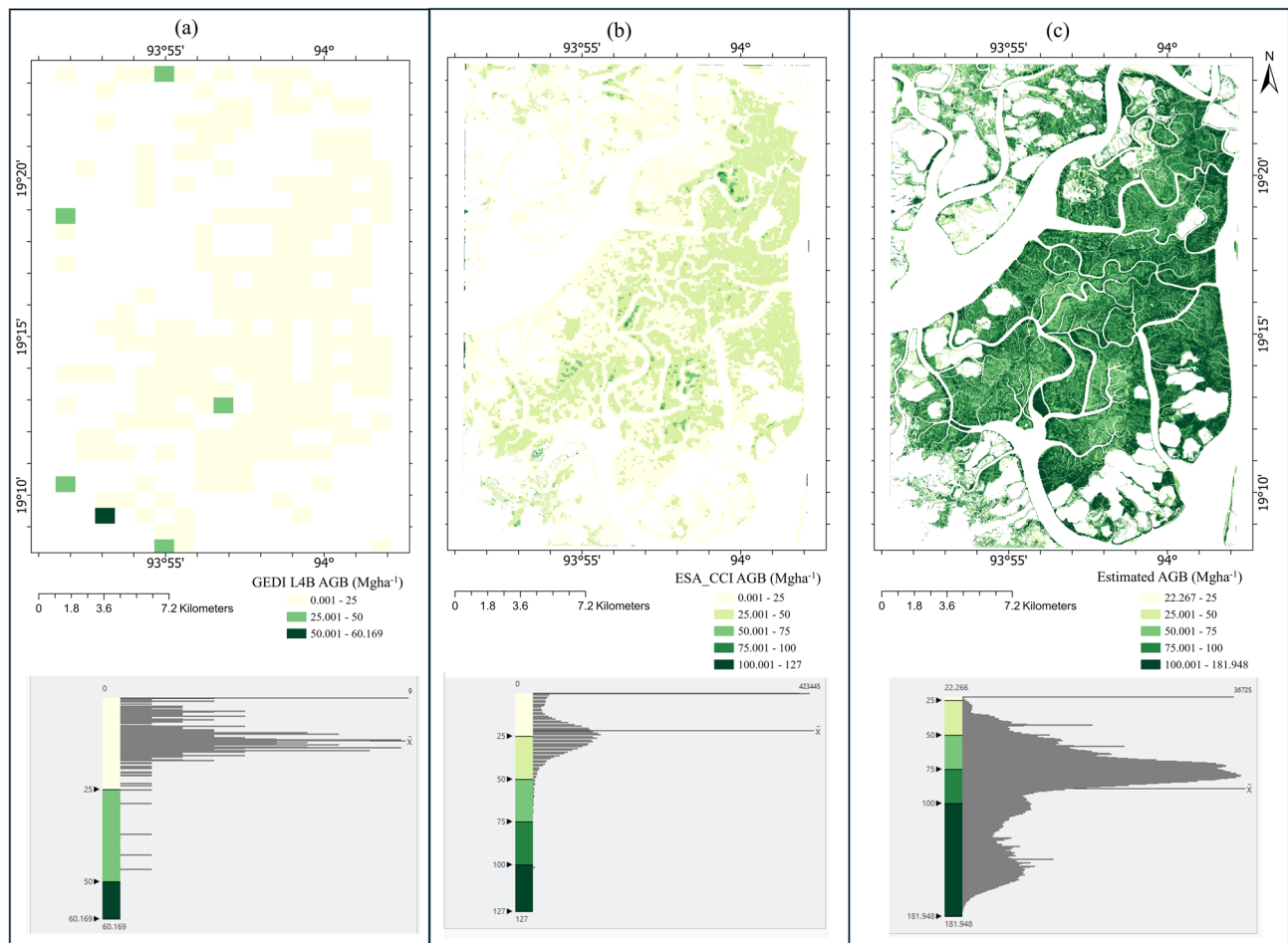
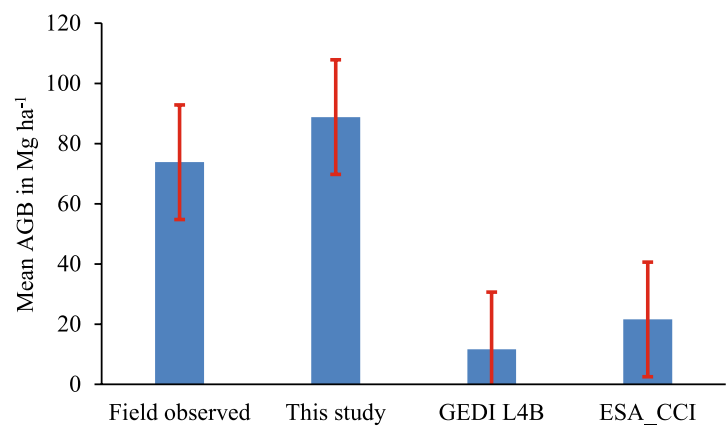


Fig. 10 Comparative AGB maps showing the spatial distribution of AGB values across the Wunbaik Mangrove Forest from three sources: **a** the GEDI_L4B dataset, **b** the ESA_CCI dataset, and **c** AGB estimates derived from this study using the RF model with multisource remote sensing data

Fig. 11 Comparison of mean AGB values between field-observed data, estimates from this study, and predictions from the GEDI_L4B and ESA_CCI datasets, illustrating the differences in AGB estimation accuracy among the sources



4.2 Random forest model for AGB estimation

In the case of the machine learning model, the results of this study demonstrated that RF consistently outperformed SVM and XGBoost in mangrove AGB estimation. SVM exhibited a notably lower R^2 and higher RMSE, indicating weak predictive performance. The linear equation of SVM shows a very weak relationship between the predicted and actual

AGB values, with a small slope that corresponds to a low R^2 and high RMSE, demonstrating that SVM struggled to explain the variability in the actual data. XGBoost, while showing a better slope than RF and SVM, had a negative R^2 and high RMSE, suggesting poorer overall performance, possibly due to overfitting or an inability to generalize to unseen data. The superior performance of RF was evidenced by a lower RMSE, indicating higher predictive accuracy and higher R^2 values, suggesting better goodness of fit between the predicted and field-observed AGB values. Since the performance of the RF model could be improved by selecting more highly correlated features, the ability of the model to capture complex, nonlinear relationships effectively between predictor variables and AGB likely contributed to its superior performance. Furthermore, the feature importance analysis provided by RF offered valuable insights into the relative importance of predictor variables in AGB estimation, aiding in the interpretation of model results and informing forest management decisions.

As a result, the RF model became a promising choice for AGB estimation in this study because of its superior performance, reasonable balance between the predicted and field AGB values, robustness, and ability to handle high-dimensional data. The findings of this study are consistent with those of some studies in which RF was used for mangrove AGB estimation. AGB estimation of dense and sparse mangroves in the Philippines was also explored using Sentinel-1 data with an RF model [64]. They reported that the RF model using the most important variable through FIV ranking could produce the highest R^2 and lowest RMSE among different feature combinations of Sentinel-1 data. A research revealed that the RF model with different feature combinations of SPOTS satellite images is capable of accurately estimating mangrove AGB in Vietnam [65]. However, with the rapid development of remote sensing and computer science, future research could explore further enhancements to Random Forest models, such as ensemble methods or model refinement techniques, to improve AGB estimation accuracy and robustness. Additionally, the integration of Random Forests with other machine learning techniques may offer synergistic benefits for AGB estimation across diverse forest ecosystems, ultimately contributing to more effective forest management and conservation strategies.

4.3 Regional mangrove AGB information for the national blue carbon strategy

As one of the largest mangrove existing countries, mangrove AGB information of Myanmar plays a crucial role in mangrove conservation and in combating global climate change internationally. The estimated AGB in this study ranged from 22.27 to 181.95 Mg ha^{-1} , with a mean value of 88.79 Mg ha^{-1} , and from 10.45 to 85.52 Mg C ha^{-1} , with a mean value of 41.73 Mg C ha^{-1} . Reviewing other field inventory-based AGB studies in Myanmar, a common allometric equation was developed for AGB estimation of 6-year-old mangrove plantations of *Avicenia marina* (Am), *Avicenia officinalis* (Ao) and *Sonneratia apetala* (Sa), and 7 years of protected naturally regenerated stands after regeneration improved felling operations in the Ayarwaddy region, Myanmar [10]. Their results revealed that the estimated mangrove AGB during natural regeneration is 70 Mg ha^{-1} , followed by Sa (69 Mg ha^{-1}), Am (25 Mg ha^{-1}), and Ao (21 Mg ha^{-1}). Field plot-based AGB estimation of natural and young planted mangroves in the Ayarwaddy, Myanmar, revealed that the estimated mean AGB of natural mangroves was 241.56 Mg ha^{-1} [11] and 100.34 Mg ha^{-1} for 3- to 5-year-old young plantations [12]. A research also revealed that the mangrove AGB at three different sites, i.e., disused ponds, young plantations, and natural mangroves, varies by 19.7, 27.4, and 121 Mg C ha^{-1} , respectively [66]. NFI data from the Forest Department, Myanmar, were used to estimate two mangrove regions: the Ayarwaddy and the Tanintharyi. According to their results, the ranges of mangrove AGB are 0.25–181.78 Mg ha^{-1} in the Ayarwaddy and 3.16–273.67 Mg ha^{-1} in the Tanintharyi [13]. Therefore, the AGB values of mangroves in Myanmar vary depending on the region, growth conditions, environmental factors, and anthropogenic factors.

With respect to the remote sensing approach, the mean AGB estimation of Southeast Asian mangroves using ALOS PALSAR was 140 Mg ha^{-1} [63]. For the AGB estimation of mangrove forests in the Philippines was also estimated using remotely sensed Interferometric Synthetic Aperture Radar (IfSAR) datasets and Sentinel-1 images, the predicted AGB of the whole mangrove forest ranged from 73.37 Mg ha^{-1} to 225.51 Mg ha^{-1} , with an average of 140.2 Mg ha^{-1} . [67]. As mentioned in Sect. 3.1, a major class of tree DBH in this study was 1–5 cm, while the maximum was 39.4 cm. Fifty out of a total of 88 plots are located at the recovering sites from human disturbances. Moreover, among the three main mangrove regions in Myanmar, mangroves in the Rakhine region experienced severe deforestation due to a lack of electricity, job opportunities, and inaccessible resource management [25]. Therefore, it was assumed that the predicted mean AGB information of this study was lower than that of other AGB estimations but slightly higher than the field AGB value of this study, as mentioned in Fig. 11.

When the AGB estimates of this study were compared with those of the GEDI L4B and ESA CCI datasets, the significant underestimation in the GEDI L4B and ESA CCI datasets highlighted the differences between the global-scale

and local-scale estimates. This comparison demonstrated the potential for enhancing AGB information when models are tailored to local conditions and incorporate finer-scale, site-specific data. Additionally, the AGB map produced in this study offered a detailed 10 m resolution, which was significantly finer than the very coarse 1 km resolution of the GEDI L4B dataset, which does not cover the entire study area. This coarse resolution likely contributes to its significant underestimation of AGB values. Despite offering a better spatial resolution than GEDI L4B does, the ESA CCI dataset also shows substantial underestimation. These factors highlighted the improvement and reliability of the AGB prediction in this study, which utilized higher-resolution data and provided a more detailed and accurate assessment of the Wunbaik Mangrove Area's biomass. Despite slightly lower mean AGB values than those reported in other studies, this study revealed improvements over two global datasets, GEDI L4B and ESA CCI. The results highlight the importance of using multisource high-resolution data and field-based measurements tailored to local conditions for accurate AGB estimation. As a first attempt at mangrove AGB estimation using Remote Sensing data in Myanmar, this study could contribute invaluable insights into the national blue carbon strategy as a benchmark of AGB information for the Wunbaik mangrove area. Future studies should develop an allometric equation that is suitable for mangroves in Myanmar. Additionally, increasing the number of AGB plots enhances the robustness of the data. A combination of field-referenced AGB using a developed allometric formula and state-of-the-art remote sensing techniques could increase the overall carbon richness of mangroves in Myanmar toward a national blue carbon strategy.

5 Conclusion

This study examined an AGB estimation method for the Wunbaik Mangrove Forest using machine learning with multisource remote sensing data. We demonstrated the feasibility and effectiveness of integrating Sentinel-2, Sentinel-1, ALOS-PALSAR HH and HV polarizations and the Canopy Height Model for mangrove AGB estimation. The model selection results revealed that the RF model is the most suitable model for Wunbaik mangrove AGB prediction. Using highly correlated feature selection, the RF model was leveraged, with a higher performance of R^2 : 0.48 and RMSE: 28.22 Mg ha⁻¹. The AGB of the Wunbaik mangrove was estimated to be within the range of 22.27–181.95 Mg ha⁻¹, with a mean value of 88.79 Mg ha⁻¹. The Wunbaik mangrove ecosystem holds a total of approximately 104.3 million tonnes of above ground carbon stock, with values ranging from 11.11 to 85.51 Mg ha⁻¹. The AGB and AGC information of this study would be an invaluable benchmark if the government could consider the blue carbon project in collaboration with international organizations. Future studies can also focus effectively on developing machine learning and remote sensing approaches using advanced models and sensors.

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Data availability The remote sensing data used in this study are available on their websites, as mentioned in the related section. Other data presented in the study are available on request from the first author.

Declarations

Competing interests The authors declare no competing interests.

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